

Instructions

- Answer all eight questions. Show sufficient working for method marks.
- A calculator may be used unless a question states otherwise.
- Write clearly in the spaces provided. Additional pages may be attached.

1

Decompose into partial fractions, stating the form you assume before finding the coefficients.

[6 marks]

$$\frac{5x + 1}{(x - 1)(x + 2)}$$

2

Integrate by parts, stating your choice of u and dv .

[5 marks]

$$\int x e^x dx$$

3

Determine whether the improper integral converges, and evaluate it if it does.

[5 marks]

$$\int_1^{\infty} \frac{1}{x^2} dx$$

4

Find the stationary points and classify each using the second derivative.

[6 marks]

$$f(x) = x^3 - 3x^2 + 4$$

5

Use the chain rule to differentiate, then evaluate at $x=0$.

[5 marks]

$$y = \sin(2x^2 + x)$$

6

Prove by induction that the statement holds for all positive integers n .

[6 marks]

$$\sum_{r=1}^n r = \frac{n(n+1)}{2}$$

7

A curve is defined implicitly. Find dy/dx in terms of x and y .

[5 marks]

$$x^2 + xy + y^2 = 7$$

8

Find the area enclosed between the two curves.

[6 marks]

$$y = x^2, \quad y = 2x$$
